

# Supplementary Material for *The Fidelity–Utility Paradox in Surrogate-Based Reinforcement Learning: A Role-Separated Hybrid Methodology in an HVAC Control Case Study*

## Supplementary Material

This document provides technical details that support, but are not required to follow, the main article; the main article reports all numerical values needed to evaluate the central fidelity–utility paradox. It collects the full nomenclature, implementation formulas, calibration and hyperparameter tables, seed-robustness and per-regime summaries, and the supporting diagnostics for each part of the study. The full project codebase is not publicly released at this stage; the building emulators are provided by the open-source BOPTEST framework (Blum et al., 2021; <https://github.com/ibpsa/project1-boptest>) and are deployed as a Docker container as described in the run-time-environment section below. All schematic figures are conceptual diagrams and are not generated from numerical datasets unless stated otherwise.

**How to read this Supplement.** It is organised in three parts that mirror the main article (predictive validation; controller utility and mechanism; and transfer boundary), followed by a section of implementation-level formulas.

**Notation (read first).** Throughout this Supplementary Material the legacy implementation labels **v3** and **v3.5** denote **BB** (the black-box surrogate) and **GB** (the grey-box surrogate) of the main article, respectively. Figure and table labels keep the legacy **v3/v3.5** names for consistency with the experimental records; all main-text prose uses BB/GB.

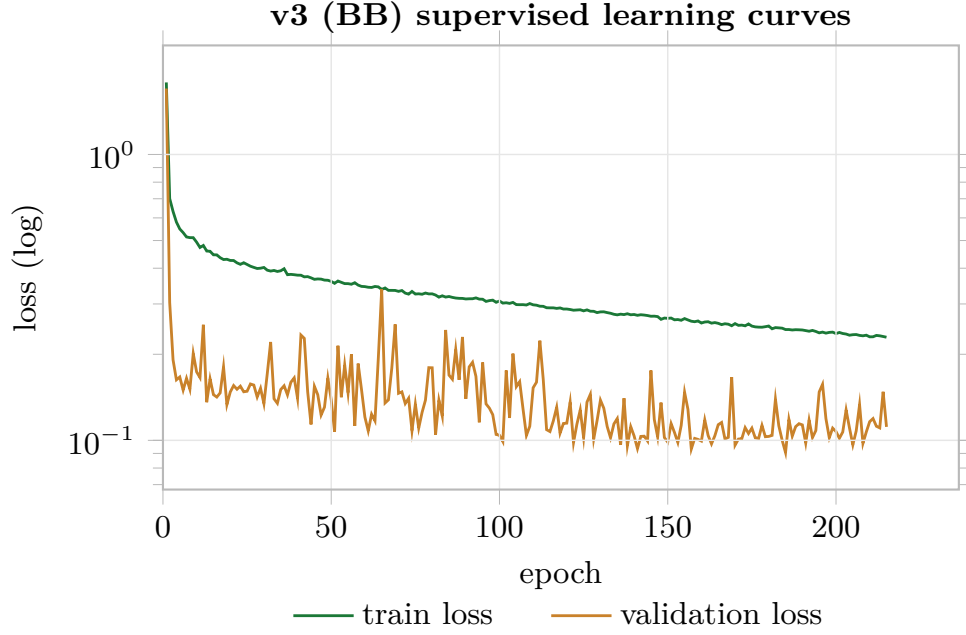
## Supplementary Nomenclature

| Symbol                                    | Unit              | Meaning                                             |
|-------------------------------------------|-------------------|-----------------------------------------------------|
| <i>Thermophysical quantities</i>          |                   |                                                     |
| $T_{\text{zone}}$                         | °C                | zone air temperature                                |
| $T_{\text{amb}}$                          | °C                | ambient air temperature                             |
| $T_{\text{sup}}$                          | °C                | supply-air temperature command                      |
| $C_{\text{zon}}$                          | J K <sup>−1</sup> | zone thermal capacitance                            |
| $R$                                       | K W <sup>−1</sup> | lumped envelope thermal resistance                  |
| $\eta$                                    | –                 | heating power conversion efficiency                 |
| $\dot{Q}$                                 | W                 | net zone heat flow                                  |
| $P$                                       | W                 | total HVAC electrical power                         |
| $\Delta t$                                | s                 | control / integration step (900 s)                  |
| <i>Reinforcement-learning formulation</i> |                   |                                                     |
| $s_t, a_t, o_t, d_t$                      | –                 | state, action, observation, disturbance at step $t$ |
| $\Phi$                                    | –                 | one-step plant (emulator) transition map            |

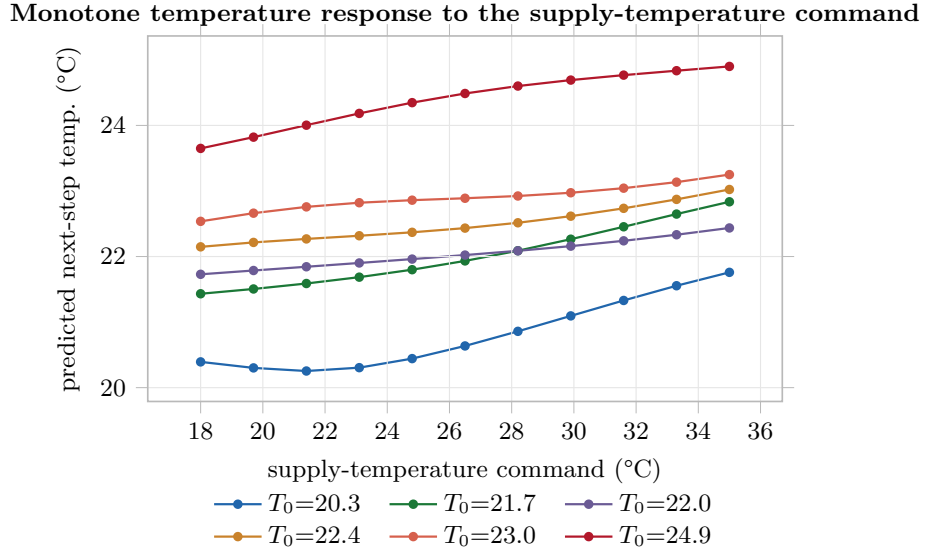
| Symbol                                           | Unit | Meaning                                                                       |
|--------------------------------------------------|------|-------------------------------------------------------------------------------|
| $f_{\theta}, g_{\phi}$                           | –    | black-box surrogate (v3) and frozen physical twin (v3.5)                      |
| $r_t$                                            | –    | per-step (shaped) reward                                                      |
| $\tilde{r}_t$                                    | –    | hybrid reward with disagreement censor                                        |
| $J(\pi)$                                         | –    | expected discounted return                                                    |
| $\gamma$                                         | –    | discount factor                                                               |
| $w_c, w_e, w_s$                                  | –    | comfort / energy / safety preference weights                                  |
| $\hat{A}_t$                                      | –    | generalized advantage estimate                                                |
| $\rho_t$                                         | –    | PPO probability ratio                                                         |
| $\epsilon$                                       | –    | PPO clipping parameter                                                        |
| $\delta$                                         | –    | surrogate–twin disagreement penalty                                           |
| $\lambda_T, \lambda_P$                           | –    | hybrid temperature / power censor weights                                     |
| <i>Evaluation metrics</i>                        |      |                                                                               |
| $m_s$                                            | –    | operational maintenance/safety score evaluated on BOPTEST (lower is better)   |
| $r_{\text{time}}, r_{\text{sev}}$                | –    | time-in-violation and severity components of $m_s$                            |
| $m_{s,\text{RL}}, m_{s,\text{PI}}$               | –    | operational score of the RL and built-in PI controller                        |
| $\text{RMSE}_T$                                  | °C   | live closed-loop zone-temperature RMSE                                        |
| $\text{CV}(\text{RMSE})$                         | %    | coefficient of variation of the RMSE                                          |
| NMBE                                             | %    | normalized mean bias error                                                    |
| $R^2$                                            | –    | coefficient of determination                                                  |
| Violation                                        | %    | fraction of steps outside the comfort band                                    |
| CV                                               | –    | coefficient of variation (std/mean) over seeds                                |
| $g_a$                                            | –    | $L_2$ action-gap norm (surrogate vs live)                                     |
| $\Delta m_s$                                     | –    | live-minus-surrogate $m_s$ transfer gap                                       |
| <i>Transferability (testcase <math>k</math>)</i> |      |                                                                               |
| $\tau$                                           | –    | pre-specified pass threshold $1.25 m_{s,\text{PI}}$                           |
| $\Delta E$                                       | %    | energy change of the RL controller vs PI                                      |
| $G_{\text{RMSE}}$                                | %    | rollout-RMSE improvement after Stage A/B/C recalibration                      |
| $\rho_C$                                         | –    | re-identified $C_{\text{zon}}$ ratio vs the <code>bestest_air</code> baseline |
| $\mathcal{A}_k$                                  | –    | pre-specified actuator adapter                                                |
| <i>Abbreviations</i>                             |      |                                                                               |
| BOPTEST                                          | –    | Building Optimization Testing Framework                                       |
| RTE                                              | –    | (BOPTEST) Run-Time Environment                                                |
| DRL / PPO                                        | –    | deep RL / proximal policy optimization                                        |
| GAE                                              | –    | generalized advantage estimation                                              |
| HDRL                                             | –    | hierarchical deep reinforcement learning                                      |
| MORL                                             | –    | multi-objective reinforcement learning                                        |
| POMDP                                            | –    | partially observable Markov decision process                                  |
| RC / ODE                                         | –    | resistance–capacitance / ordinary differential equation                       |
| PI / RBC / MPC                                   | –    | proportional–integral / rule-based / model predictive control                 |

## Supplementary figures and tables

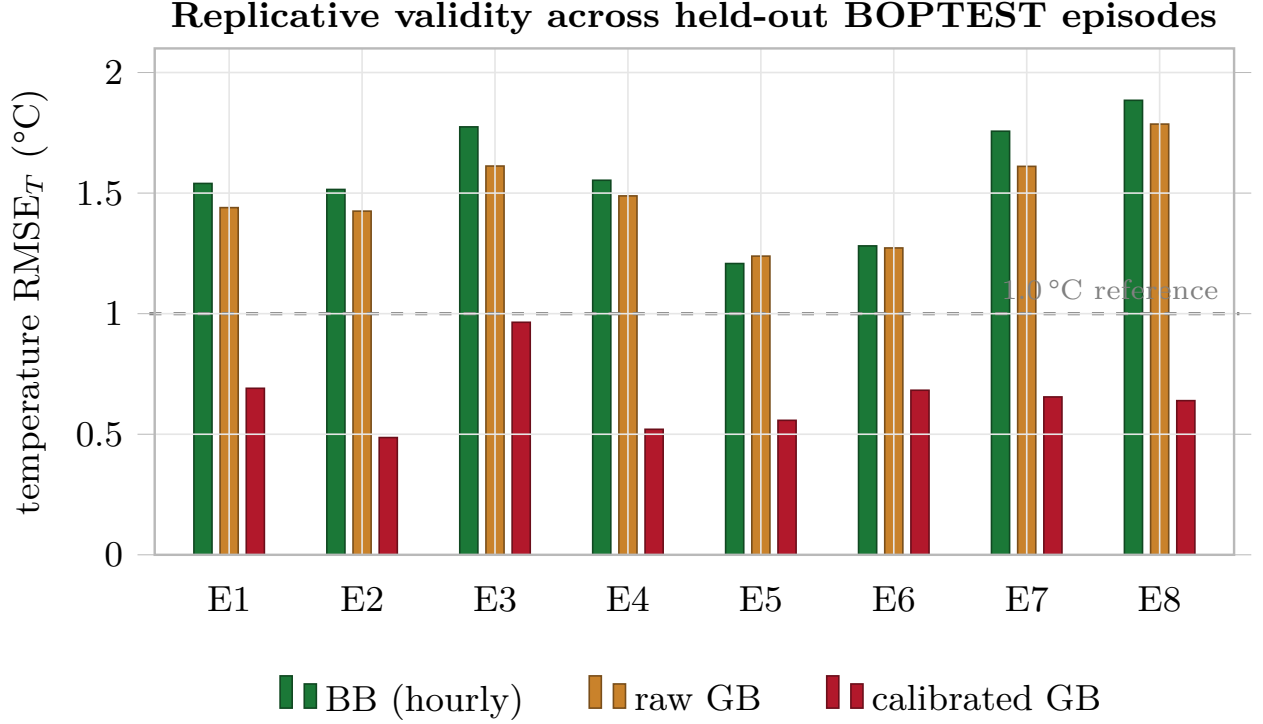
The following figures and tables provide supporting schematics, calibration and controller diagnostics, and detailed per-regime and per-seed breakdowns. They are cited from the main text (as Supplementary Figure/Table references) but are not required to follow the main argument.



**Figure S1:** v3 supervised train/validation learning curves. Early stopping at the best validation epoch precedes any validation-loss upturn, indicating no overfitting at the selected checkpoint.



**Figure S2:** Physical-consistency audit: predicted next-step zone temperature versus the supply-temperature command for representative initial states. The response has the correct control sign and is predominantly increasing (mean Spearman 0.91), with small departures from strict monotonicity contributed by the neural residual head.



**Figure S3:** Replicative validity across the eight held-out BOPTEST episodes used by the predictive-validation prepared-rollout results. The dashed line marks a 1 °C engineering reference.

**Table S1:** Predictive-validation data corpora referenced by the experimental protocol.

| Dataset                   | Rows   | Step (s) | Policy mix                                           | Scenario mix                          |
|---------------------------|--------|----------|------------------------------------------------------|---------------------------------------|
| v3 canonical corpus       | 51,200 | 3600     | cool, heat, mixed, random                            | autumn, spring, summer, winter        |
| v3.5 calibration corpus   | 10,744 | 900      | hdrl, thermostatic                                   | peak_heat_window, typical_heat_window |
| 15-min exploration corpus | 48,384 | 900      | cool, heat, mixed, pulse, random, thermostatic_noise | autumn, spring, summer, winter        |

**Table S2:** Architecture and calibration summary for the two surrogate roles.

| Surrogate       | Role                                   | Architecture / physics                                                                                                                | Corpus                             | Training or calibration protocol                                                              |
|-----------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------|
| v3 direct-TSup  | Fast control-oriented rollout dynamics | 8D dual-head MLP: HeatFlowNetV2 (7,105 params, LayerNorm/Tanh/residual), PowerNetV2 (1,377 params, Tanh/Softplus); total 8,482 params | 51,200 hourly transitions          | AdamW, lr=1e-3, batch=256, epochs=500                                                         |
| v3.5 calibrated | Physics-informed predictive twin       | RC-NeuralODE backbone with positive C <sub>zon</sub> reparameterization and residual temperature/power heads                          | 10,744 prepared 15-min transitions | Stage A latency/bias; Stage B 120 epochs; Stage C 60+80 epochs; C <sub>zon</sub> =4.413e5 J/K |

**Table S3:** v3 configuration from the active model class and Hou-and-Evins reporting results.

| Item             | Value                                   |
|------------------|-----------------------------------------|
| Input dimension  | 8                                       |
| Temperature head | HeatFlowNetV2, 7,105 params             |
| Power head       | PowerNetV2, 1,377 params                |
| Total parameters | 8,482                                   |
| Epoch budget     | 500                                     |
| Batch size       | 256                                     |
| Learning rate    | 1e-3                                    |
| Optimizer        | AdamW                                   |
| Early stopping   | patience=30; improvement_threshold=1e-6 |

**Table S4:** Feature scaling and physical constraints used by the surrogate interface.

| Variable          | Scaling method               | Parameters                                                               | Engineering role                                                                               |
|-------------------|------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| surrogate_t_zone  | affine min-max to [-1, 1]    | min=15 C; max=35 C                                                       | Keeps the learned thermal state within the training support used by the direct-TSup surrogate. |
| surrogate_t_amb   | affine min-max to [-1, 1]    | min=-10 C; max=40 C                                                      | Normalizes outdoor-air temperature for the heat and power heads.                               |
| hour              | cyclic sine/cosine           | $\sin(2\pi \cdot \text{hour}/24)$ ;<br>$\cos(2\pi \cdot \text{hour}/24)$ | Avoids discontinuity between hour 23 and hour 0.                                               |
| day               | cyclic sine/cosine           | $\sin(2\pi \cdot \text{day}/365)$ ;<br>$\cos(2\pi \cdot \text{day}/365)$ | Represents seasonal phase without an artificial year-end jump.                                 |
| a0_raw            | native action domain         | [-1, 1] maps to 18.35 C supply temperature                               | Preserves the PPO action coordinate while keeping physical supply-temperature bounds explicit. |
| a1_raw            | native action domain         | [-1, 1] maps to 0..1 fan command                                         | Keeps fan modulation aligned with the controller action range.                                 |
| power_head_output | Softplus multiplied by P_MAX | P_MAX=5500 W                                                             | Guarantees non-negative predicted HVAC power.                                                  |

**Table S5:** Stage A telemetry-alignment operations from the calibration-summary JSON.

| Operation                     | Estimated value           | Effect                       |
|-------------------------------|---------------------------|------------------------------|
| Latency compensation          | 1 step (15 min)           | latency-search RMSE 0.378 C  |
| Temperature bias removal      | -0.1049 C                 | median residual offset       |
| Power affine normalization    | scale -0.702, bias 1148 W | global power-channel rescale |
| Post-Stage-A temperature RMSE | 0.2383 C                  | baseline for Stage B         |
| Post-Stage-A power MAE        | 514.4 W                   | baseline for Stage C         |

**Table S6:** Stage B excitation filtering and physical identification of  $C_{\text{zon}}$ .

| Parameter           | Value                   | Notes                      |
|---------------------|-------------------------|----------------------------|
| C_zon prior         | $4.200 \times 10^5$ J/K | air volume x specific heat |
| C_zon after Stage B | $4.413 \times 10^5$ J/K | +5.1% vs prior             |
| Stage B epochs      | 120                     | episode-aware run          |
| C_zon learning rate | $10^{-3}$               | dedicated scalar LR        |
| Excitation quantile | 0.95                    | dT  threshold 0.1759       |
| Excitation rows     | 403 / 8058              | mean score 0.240 vs 0.069  |

**Table S7:** Stage A/B/C calibration summary from the canonical JSON and rollout results.

| Metric                                              | Before | After | Relative change |
|-----------------------------------------------------|--------|-------|-----------------|
| 1-step $\text{RMSE}_T$ ( $^{\circ}\text{C}$ )       | 0.384  | 0.235 | -38.8%          |
| 24 h rollout $\text{RMSE}_T$ ( $^{\circ}\text{C}$ ) | 1.466  | 0.644 | -56.1%          |
| Power MAE (W)                                       | 808    | 482   | -40.3%          |
| $C_{\text{zon}}$ ( $\times 10^5$ J/K)               | 4.200  | 4.413 | +5.1%           |

**Table S8:** Multi-horizon teacher-forced rollout fidelity from the Hou-and-Evins predictive-validity table. Cells are temperature RMSE ( $^{\circ}\text{C}$ ) at each horizon; the final column is the 24 h temperature  $R^2$  (n/a where not computed for the auxiliary baselines).

| Model               | 1 h   | 4 h   | 8 h   | 24 h  | 24 h $R_T^2$ |
|---------------------|-------|-------|-------|-------|--------------|
| v3 (hourly)         | 1.471 | 1.579 | 1.579 | 1.558 | -1.411       |
| v3 (15-min matched) | 0.875 | 0.874 | 0.873 | 0.876 | n/a          |
| raw v3.5            | 1.491 | 1.491 | 1.491 | 1.466 | n/a          |
| calibrated v3.5     | 0.321 | 0.575 | 0.638 | 0.644 | +0.588       |

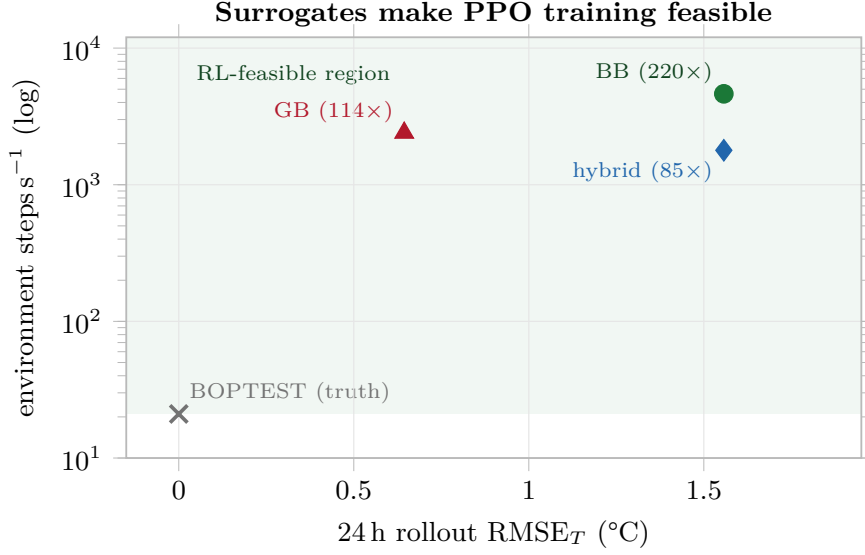
**Table S9:** Backend throughput under the 900 s control protocol from the speed-benchmark result (single CPU thread, BOPTEST reached through the same HTTP-Docker interface used in training).

| Backend               | Steps/s | Median (ms) | P95 (ms) | Speed-up       |
|-----------------------|---------|-------------|----------|----------------|
| BOPTEST RTE HTTP loop | 21.0    | 14.915      | 19.251   | 1.0 $\times$   |
| v3 surrogate          | 4,626.3 | 0.165       | 0.273    | 220.2 $\times$ |
| v3.5 calibrated       | 2,399.5 | 0.359       | 0.495    | 114.2 $\times$ |
| hybrid v3+v3.5        | 1,786.8 | 0.507       | 0.648    | 85.0 $\times$  |

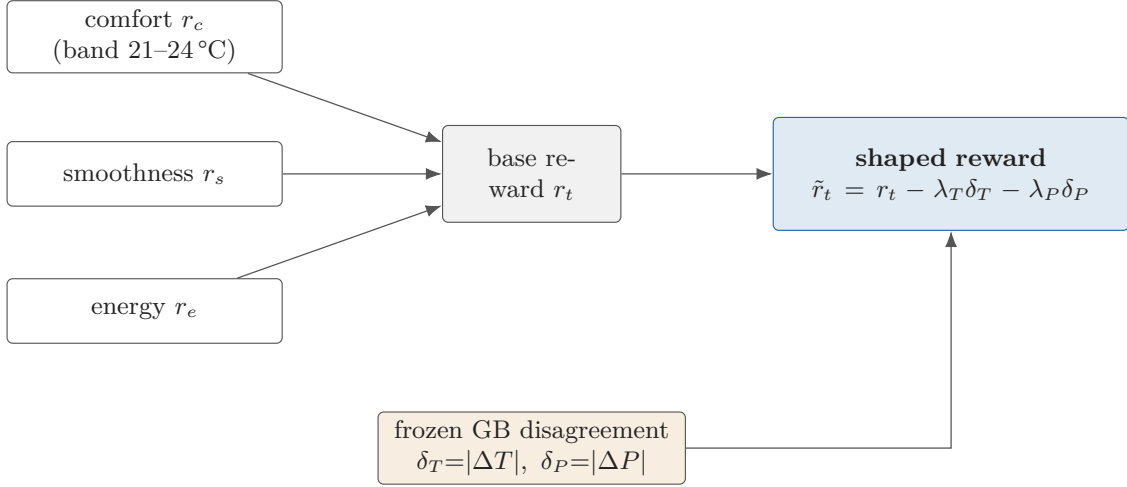


*training never touches the live emulator; the policy is frozen before deployment (zero-shot to BOPTEST)*

**Figure S4:** Controller-learning pipeline. The experiments separate the rollout model, the physical regularizer, the controller family, the observation interface, and live BOPTEST validation.

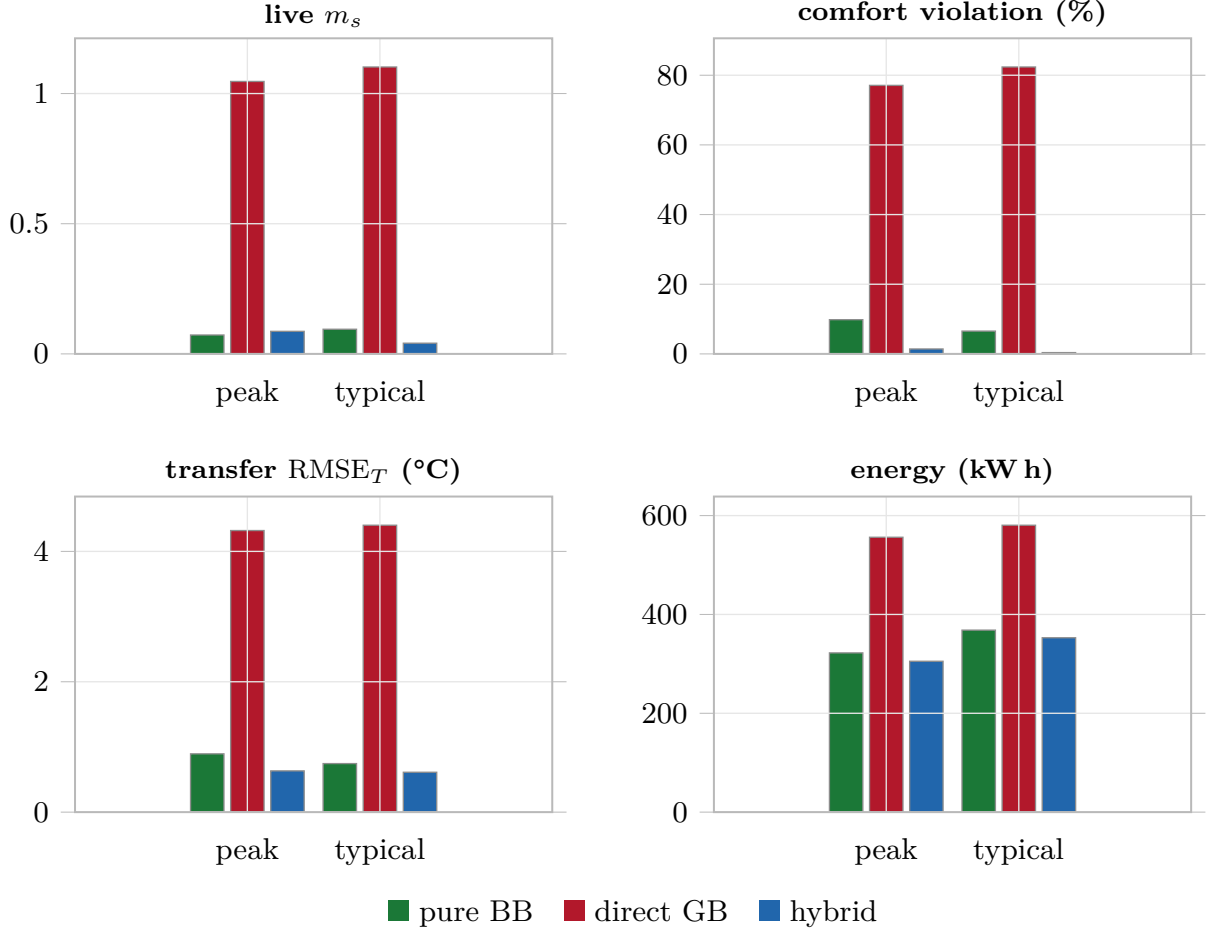


**Figure S5: Surrogates make policy-gradient training computationally feasible.** Environment throughput (env steps/s, log scale) versus 24 h rollout RMSE<sub>T</sub> for the BOPTEST emulator (ground truth) and the three in-process surrogate backends. The surrogates run 85–220× faster than the BOPTEST RTE HTTP testbed, turning a days-long PPO run into hours; the hybrid is plotted at v3’s rollout RMSE because it rolls out on v3. The plotted values combine measured throughput with 24 h rollout-RMSE estimates.

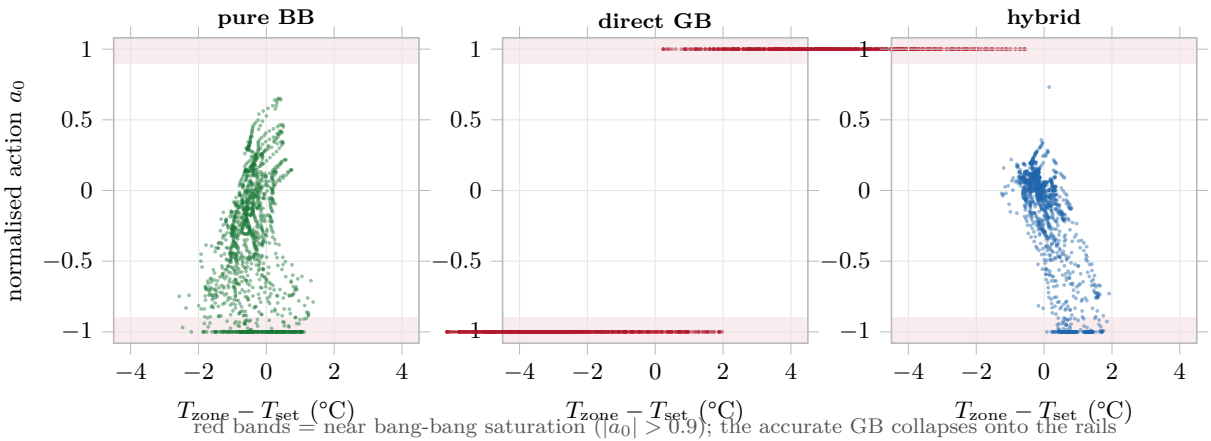


$$\lambda_T = 0.10, \lambda_P = 5 \times 10^{-5}; \text{ the GB penalty is forward-only and never enters the policy loss.}$$

**Figure S6: Hybrid reward-shaping mechanism.** v3 supplies rollout dynamics; frozen v3.5 supplies a disagreement penalty on the same state-action transition.

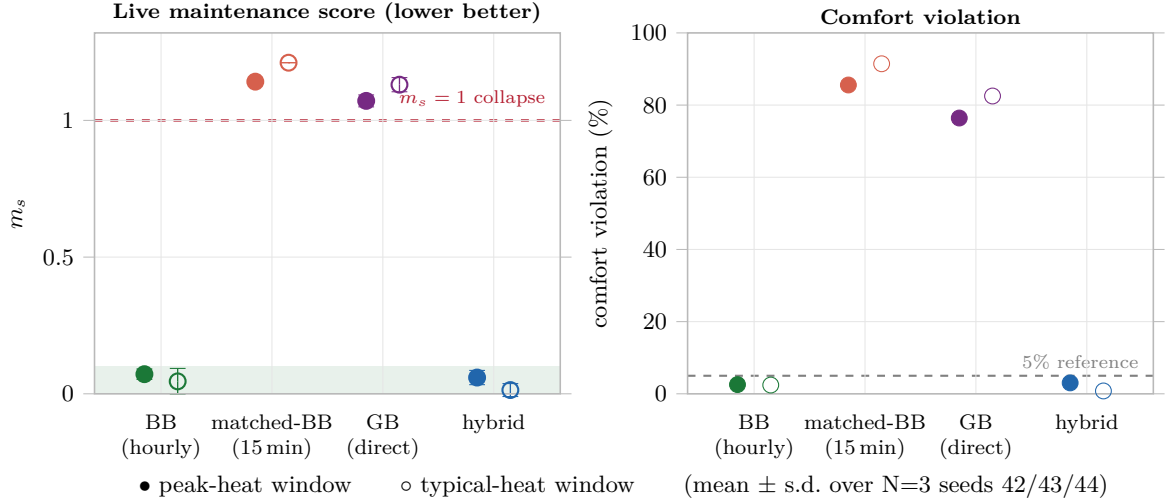


**Figure S7:** Detailed live BOPTEST KPI breakdown (per-metric bars) for pure v3, direct v3.5, and hybrid PPO. Direct v3.5 is a negative control: higher predictive fidelity does not imply a useful RL training environment. These per-metric bars give the underlying  $m_s$ , violation, RMSE, and energy behind the compact fidelity–utility scatter in the main text.

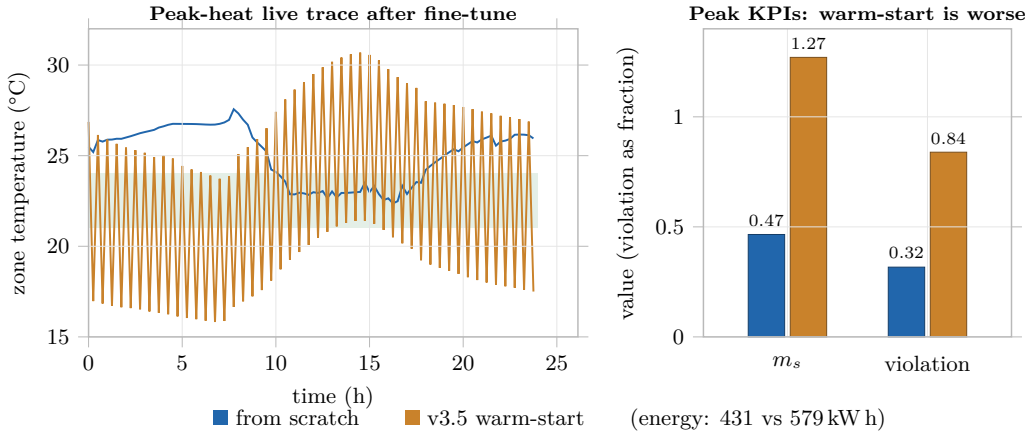


**Figure S8:** Action-density phase portrait of the three thermostatic controllers on the live BOPTEST `typical_heat_window`: empirical hexbin density of the normalised supply-temperature action  $a_0$  (one hue per backend) against the thermal error  $T_{\text{zone}} - T_{\text{set}}$ , on *identical* axes, with the marginal  $a_0$  distribution in the right strip of each panel and the measured saturation share ( $|a_0| \geq 0.9$ ). The red bands mark the actuator-saturation region. Direct v3.5 collapses onto  $a_0 = -1$  (100% saturation, a bang-bang law) whereas the usable hourly v3 and the hybrid modulate ( $\approx 19\text{--}20\%$ ).

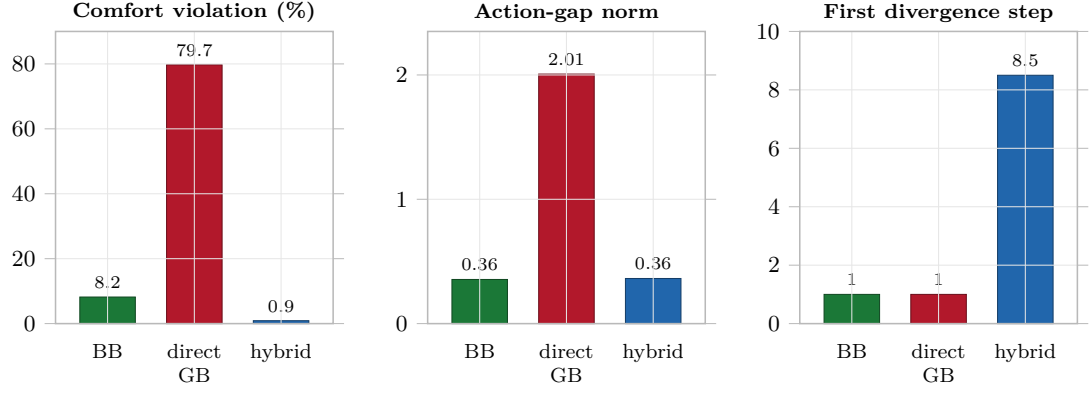




**Figure S9:** Across  $N = 3$  seeds the fidelity–utility paradox is not a seed result: the usable hourly-v3 and hybrid controllers stay below both the  $m_s = 1$  closed-loop-collapse line and the 5% comfort-violation reference on every seed, while the matched-resolution v3 and the direct v3.5 both collapse on every seed ( $m_s > 1$ ,  $> 75\%$  violation) with small seed variance. Markers are the mean over seeds {42,43,44} (filled = peak-heat window, open = typical-heat window); whiskers are  $\pm$  one standard deviation; colour encodes the controller (green = coarse v3, salmon = matched-resolution v3, purple = direct v3.5, blue = hybrid). **(left)** live maintenance score  $m_s$  (lower is better, with the usable  $m_s < 0.1$  band shaded); **(right)** comfort-violation fraction. Same data as Table S14.

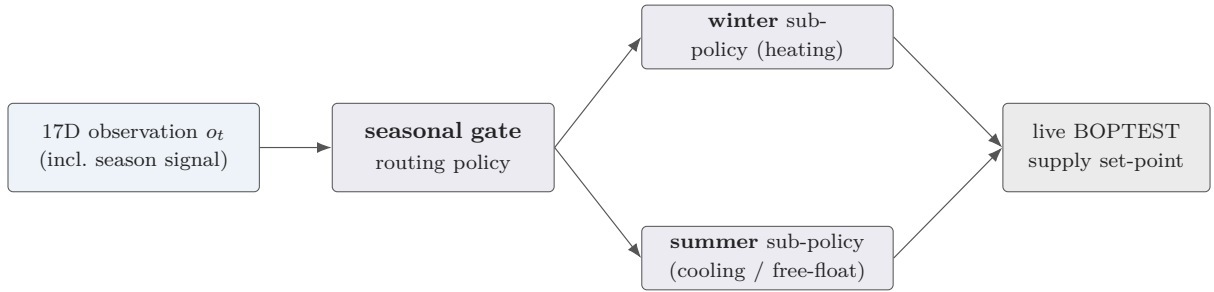


**Figure S10:** Warm-start negative control. Pretraining on direct v3.5 and then fine-tuning on the hybrid backend is inferior to training the hybrid policy from scratch.



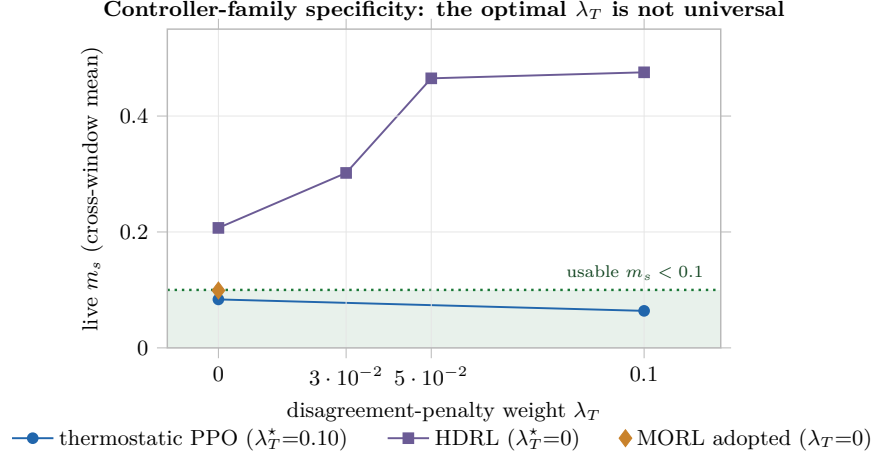
**Transfer-gap diagnostics: standalone direct-GB training saturates (bang-bang)**

**Figure S11:** Transfer-gap diagnostics. Direct v3.5 has high action-gap norm and live-surrogate mismatch; the hybrid backend suppresses both.

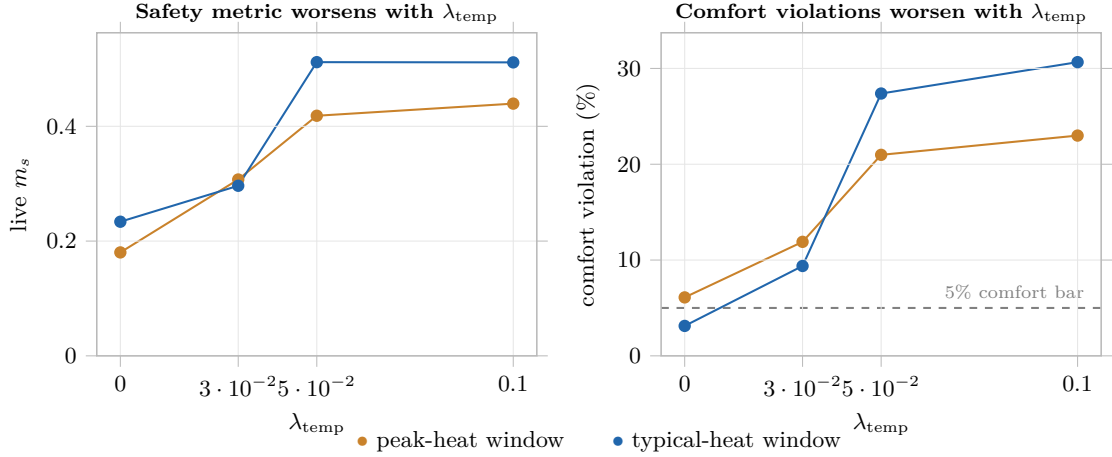


*The gate selects a season-specialised sub-policy; the disagreement penalty  $\lambda_T$  that helps thermostatic PPO does not transfer to this family (controller-family specific).*

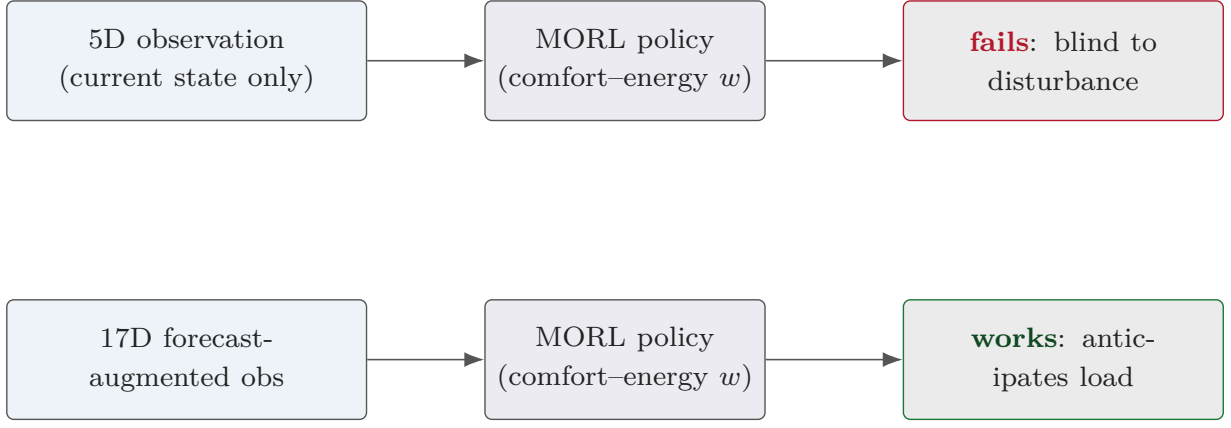
**Figure S12:** HDRL architecture: a high-level seasonal gate  $k(t)$  routes the 17D observation to one of two low-level PPO setpoint specialists (winter / summer). Because the low level is already comfort-aware, the v3.5 temperature sensor over-regularizes it.



**Figure S13: The optimal censor weight  $\lambda_T$  is controller-family specific.** Live BOPTEST maintenance score  $m_s$  (cross-window mean of the peak and typical windows) versus  $\lambda_T$  for each controller family;  $\star$  marks the per-family optimum. Thermostatic PPO improves with the censor (optimum  $\lambda_T = 0.10$ , the canonical hybrid), HDRL is best *without* it ( $\lambda_T = 0.00$ ) and degrades monotonically, and MORL adopts  $\lambda_T = 0.00$ . HDRL has a dense four-point sweep; PPO is shown at its two measured endpoints (pure v3 =  $\lambda_T 0$ , hybrid = 0.10) and MORL at its single adopted configuration. The panel compares the *location* of each optimum, not absolute  $m_s$  across families (they use different observation/action interfaces).

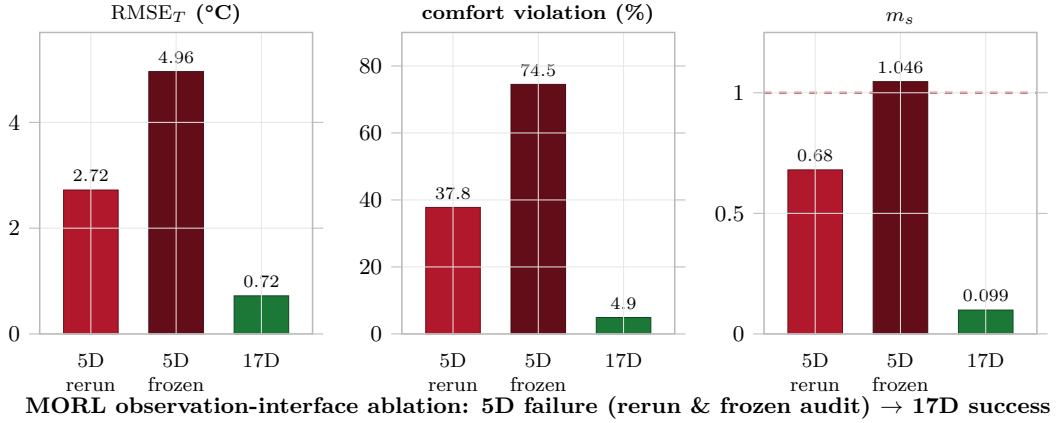


**Figure S14: HDRL  $\lambda_T$  sensitivity.** The thermostatic regularization weight does not transfer; the best HDRL policy uses no temperature-disagreement term.



*The 5D→17D observation-interface change (adding forecast context), not the reward weighting, is what makes MORL control-viable.*

**Figure S15:** MORL observation-interface schematic. The 5D instantaneous observation fails because it is blind to forecasted disturbances, whereas the 17D forecast-augmented observation makes MORL control-viable. The quantitative ablation of this schematic is in Figure S16.



**Figure S16:** MORL observation-interface ablation. The main article reports the headline 5D-versus-17D numbers in its controller-family boundary-test paragraph; this supplementary plot includes the current-code 5D rerun, the frozen historical 5D audit result, and the 17D power-only recovery.

**MORL 17D yearly seasonal validation**

|               |      |      |      |      |     |      |      |      |      |      |      |      |
|---------------|------|------|------|------|-----|------|------|------|------|------|------|------|
| violation (%) | 5.4  | 2.5  | 3.7  | 0.5  | 0   | 0.3  | 2.2  | 2.3  | 8.3  | 11.7 | 13.2 | 8.3  |
| energy (kW h) | 403  | 397  | 320  | 250  | 173 | 137  | 125  | 134  | 167  | 193  | 264  | 419  |
| live $m_s$    | 0.14 | 0.06 | 0.08 | 0.02 | 0   | 0.02 | 0.08 | 0.08 | 0.16 | 0.2  | 0.2  | 0.14 |
|               | Jan  | Feb  | Mar  | Apr  | May | Jun  | Jul  | Aug  | Sep  | Oct  | Nov  | Dec  |

**Figure S17:** MORL seasonal performance heatmap under the 17D interface. The figure shows monthly structure but does not support the N=3 mechanism claim after N=5 extension.

**MORL seasonal variance after N=5 falsification (across-seed  $m_s$  std)**

|                               |       |       |       |       |       |       |       |       |       |       |       |       |
|-------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Practical<br>$w=(0.75, 0.25)$ | 0.124 | 0.188 | 0.106 | 0.171 | 0.091 | 0.075 | 0.076 | 0.089 | 0.023 | 0.058 | 0.121 | 0.181 |
| Neutral<br>$w=(0.50, 0.50)$   | 0.114 | 0.143 | 0.089 | 0.136 | 0.098 | 0.057 | 0.057 | 0.09  | 0.033 | 0.099 | 0.132 | 0.154 |
|                               | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |

**Figure S18:** Post-N=5 seasonal variance diagnostic. The earlier seasonal-inversion hypothesis is reported as falsified, not retained as a positive mechanism claim.**Table S10:** Targeted-window and yearly scenario definitions.

| Scenario            | Day idx   | Start (s) | Dur. (d) | Ambient ( $^{\circ}\text{C}$ ) | Role                                    |
|---------------------|-----------|-----------|----------|--------------------------------|-----------------------------------------|
| peak_heat_window    | 3         | 259200    | 14       | -24.4                          | January coldest, heating stress test    |
| typical_heat_window | 37        | 3196800   | 14       | 2.4                            | February moderate, deployment-realistic |
| yearly evaluation   | 12 months | —         | 14/mo    | varied                         | MORL + PI yearly summary                |

**Table S11:** PPO training configuration used in the controller-learning experiments.

| Parameter       | Thermostatic PPO     | HDRL                 | MORL pretrain        | MORL finetune        |
|-----------------|----------------------|----------------------|----------------------|----------------------|
| Algorithm       | PPO, MlpPolicy       | PPO, MlpPolicy       | PPO, MlpPolicy       | PPO load+continue    |
| Learning rate   | $3.0 \times 10^{-4}$ | $3.0 \times 10^{-4}$ | $3.0 \times 10^{-4}$ | $1.0 \times 10^{-4}$ |
| n_steps         | 1024                 | 1024                 | 2048                 | inherited            |
| batch_size      | 4096                 | 2048                 | 64                   | inherited            |
| n_epochs        | 10                   | 10                   | 10                   | 10                   |
| $\gamma$        | 0.99                 | 0.99                 | 0.99                 | 0.99                 |
| GAE $\lambda$   | 0.95                 | 0.95                 | 0.95                 | 0.95                 |
| Clip range      | 0.20                 | 0.20                 | 0.20                 | 0.20                 |
| Total timesteps | 10M                  | 12M                  | 2M                   | 100k                 |
| Seed            | 42                   | 42                   | 42                   | 42–46 (N=5)          |

**Table S12:** Extended 17D observation feature groups.

| Feature group          | Dim. | Contents                                                                                         |
|------------------------|------|--------------------------------------------------------------------------------------------------|
| Physical state (basic) | 5    | $T_{\text{zone}}$ , $\text{CO}_2$ , clipped-log power, prev. $T_{\text{sup}}$ , $T_{\text{amb}}$ |
| Cyclic time            | 4    | hour and day sine/cosine encodings                                                               |
| Ambient forecast       | 5    | $T_{\text{amb}}$ at +1, +3, +6, +12, +24 h                                                       |
| Previous action        | 2    | $(a_{T_{\text{sup}}}, a_{\text{fan}})$ from last step                                            |
| Temperature delta      | 1    | causal-smoothed $\Delta T_{\text{zone}}$                                                         |
| Total (extended)       | 17   | obs_mode = extended                                                                              |

The shaped training reward is  $r_t = -(w_c c_t + w_e e_t)$ , with a piecewise-linear, asymmetric comfort term

$$r_{\text{comfort},t} = \begin{cases} -w_u (T_\ell - T_t) + b_{\text{heat}} & T_t < T_\ell, \\ +\beta & T_\ell + \delta \leq T_t \leq T_h - \delta, \\ -w_o (T_t - T_h) + b_{\text{cool}} & T_t > T_h, \end{cases} \quad (1)$$

and an energy term  $r_{\text{energy},t} = -\eta P_t$  ( $\eta = 2 \times 10^{-4}$ ); the cold/hot weights are ambient-dependent and fixed a priori. The parameters are listed below.

**Table S13:** Reward-shaping parameters.

| Component                      | Value                                | Role                               |
|--------------------------------|--------------------------------------|------------------------------------|
| Comfort band                   | 21.0 / 24.0 C                        | comfort interval (band_low/high)   |
| Comfort deadband               | 0.50 C                               | soft margin around band edges      |
| In-band bonus                  | 0.050/step                           | reward for staying inside the band |
| Undershoot / overshoot weight  | 1.15 / 1.15                          | generic violation penalties        |
| Cold-ambient asymmetry         | 1.60 (amb < 8.0 C)                   | extra cold-undershoot penalty      |
| Hot-ambient asymmetry          | 1.80 (amb > 24.0 C)                  | extra hot-overshoot penalty        |
| Heating action bonus           | 0.04 ( $T_{\text{sup}} \geq 29.0$ C) | anti-degenerate shaping            |
| Cooling action bonus           | 0.06 ( $T_{\text{sup}} \leq 21.0$ C) | anti-degenerate shaping            |
| MORL weights ( $w_c/w_e/w_s$ ) | 0.80 / 0.20 / 0.00                   | canonical scalarization            |
| Energy scale                   | $2e - 4$ (W $\rightarrow$ reward)    | energy-to-reward conversion        |

**Table S14:** Seed robustness ( $N = 3$  seeds {42,43,44}) of the thermostatic controllers on the live BOPTEST targeted windows; mean  $\pm$  standard deviation. The usable controllers (pure v3, hybrid) stay below the 5% comfort-violation bar on every seed, while the collapse controls (matched-resolution v3, direct v3.5, and the  $\Delta t$ -rescaled hourly v3) exceed  $m_s = 1$  on every seed with small seed variance. The  $\Delta t$ -rescaled control keeps the hourly v3 but rescales its increment to the 900 s runtime step; its collapse shows the hourly BB’s usability rests on its large coarse-scale increment.

| Controller                  | Window  | $N$ | $m_s$ (mean $\pm$ std) | Violation % (mean $\pm$ std) |
|-----------------------------|---------|-----|------------------------|------------------------------|
| pure v3                     | peak    | 3   | $0.072 \pm 0.019$      | $2.6 \pm 1.2$                |
| pure v3                     | typical | 3   | $0.045 \pm 0.047$      | $2.4 \pm 2.2$                |
| matched v3 (15-min)         | peak    | 3   | $1.142 \pm 0.000$      | $85.6 \pm 0.0$               |
| matched v3 (15-min)         | typical | 3   | $1.211 \pm 0.000$      | $91.4 \pm 0.0$               |
| direct v3.5                 | peak    | 3   | $1.071 \pm 0.022$      | $76.4 \pm 0.7$               |
| direct v3.5                 | typical | 3   | $1.131 \pm 0.026$      | $82.5 \pm 0.7$               |
| $\Delta t$ -rescaled v3     | peak    | 3   | $1.187 \pm 0.010$      | $83.2 \pm 0.7$               |
| $\Delta t$ -rescaled v3     | typical | 3   | $1.215 \pm 0.006$      | $86.0 \pm 0.3$               |
| hybrid ( $\lambda_T=0.10$ ) | peak    | 3   | $0.059 \pm 0.026$      | $3.0 \pm 1.6$                |
| hybrid ( $\lambda_T=0.10$ ) | typical | 3   | $0.014 \pm 0.024$      | $0.8 \pm 1.4$                |

**Table S15:** Measured response-surface smoothness of the three single-model backends, probed directly on each surrogate’s one-step action map (no controller, no BOPTEST). The scale-free relative roughness (curvature normalised by slope) is independent of the step length and tracks closed-loop usability: the usable hourly v3 is the smoothest; both collapsed backends are several times rougher.

| Surrogate backend     | Slope $ \partial \hat{T} / \partial a_0 $ | Rel. roughness | vs. v3       | Closed-loop role |
|-----------------------|-------------------------------------------|----------------|--------------|------------------|
| canonical v3 (hourly) | 3.368                                     | 0.169          | 1.0 (ref)    | usable           |
| matched v3 (15-min)   | 0.272                                     | 1.578          | 9.4 $\times$ | collapse         |
| calibrated v3.5       | 0.467                                     | 1.338          | 7.9 $\times$ | collapse         |

**Table S16:** Thermostatic hybrid  $\lambda_T$  sweep *design* (experimental protocol §5;  $\lambda_P = 5 \times 10^{-5}$  throughout); the mid point is the retained canonical. Of these pre-specified brackets, the live-BOPTEST two-window evaluation reported in the paper covers thermostatic PPO at  $\lambda_T = 0.10$  (canonical hybrid) and  $\lambda_T = 0$  (pure BB), so the controller-family  $\lambda_T$  figure shows PPO at these two measured endpoints; the 0.05 and 0.15 brackets were not carried to full live two-window evaluation. The dense four-point  $\lambda_T$  sweep is the HDRL family, not PPO.

| $\lambda_T$ | Tag         | Role                                                           |
|-------------|-------------|----------------------------------------------------------------|
| 0.05        | hybrid_l005 | weaker censor (sweep bracket)                                  |
| 0.10        | hybrid_l010 | <b>canonical</b> (retained; $m_s = 0.087$ peak, 0.041 typical) |
| 0.15        | hybrid_l015 | stronger censor (sweep bracket)                                |

**Table S17:** Direct-v3.5 warm-start utility.

| Mode                   | Scenario | $m_s$ | Violation (%) |
|------------------------|----------|-------|---------------|
| scratch (random init)  | peak     | 0.465 | 31.70         |
| scratch (random init)  | typical  | 0.578 | 45.54         |
| warm-start (from v3.5) | peak     | 1.270 | 83.93         |
| warm-start (from v3.5) | typical  | 1.289 | 86.83         |

**Table S18:** Transfer diagnostics across three backends.  $\Delta m_s = \text{surrogate} - \text{live}$  (negative  $\Rightarrow$  surrogate optimistic); Temp gap is the live closed-loop  $\text{RMSE}_T$ .

| Variant                   | Scenario | Temp gap ( $^{\circ}\text{C}$ ) | $\Delta m_s$ | Action gap | First div. | Top driver   |
|---------------------------|----------|---------------------------------|--------------|------------|------------|--------------|
| pure v3                   | peak     | 0.894                           | -0.102       | 0.377      | 1          | p_total_norm |
| pure v3                   | typical  | 0.745                           | -0.068       | 0.333      | 1          | p_total_norm |
| hybrid $\lambda_T = 0.10$ | peak     | 0.633                           | -0.023       | 0.473      | 1          | p_total_norm |
| hybrid $\lambda_T = 0.10$ | typical  | 0.612                           | -0.021       | 0.253      | 16         | p_total_norm |
| direct v3.5               | peak     | 4.320                           | -0.886       | 2.000      | 1          | t_zone_norm  |
| direct v3.5               | typical  | 4.401                           | -1.014       | 2.014      | 1          | t_zone_norm  |

**Table S19:** HDRL sweep over  $\lambda_T$  on the targeted windows; best is  $\lambda_T = 0$ , and performance degrades monotonically as the temperature censor strengthens.

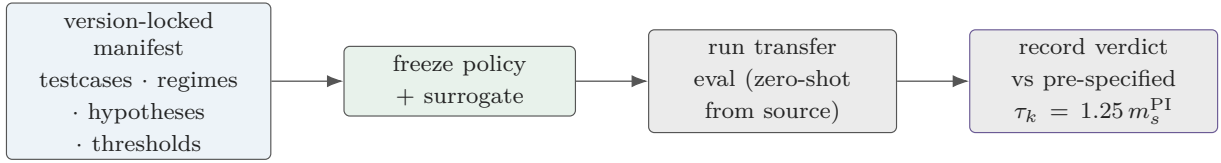
| $\lambda_T$ | Scenario | $m_s$ | Violation (%) | $\text{RMSE}_T$ ( $^{\circ}\text{C}$ ) | Energy (kWh) |
|-------------|----------|-------|---------------|----------------------------------------|--------------|
| 0.00        | peak     | 0.180 | 6.10          | 0.751                                  | 329.6        |
| 0.03        | peak     | 0.307 | 11.90         | 0.993                                  | 311.5        |
| 0.05        | peak     | 0.418 | 20.98         | 1.303                                  | 298.1        |
| 0.10        | peak     | 0.440 | 22.99         | 1.343                                  | 300.4        |
| 0.00        | typical  | 0.234 | 3.12          | 0.691                                  | 385.1        |
| 0.03        | typical  | 0.296 | 9.38          | 0.959                                  | 369.6        |
| 0.05        | typical  | 0.512 | 27.38         | 1.491                                  | 354.5        |
| 0.10        | typical  | 0.511 | 30.65         | 1.455                                  | 357.1        |

**Table S20:** MORL Pareto (seed 42) and N=5 canonical seed analysis.

| Preference / statistic | RMSE <sub>T</sub> (°C) | Violation (%) | $m_s$       | Interpretation         |
|------------------------|------------------------|---------------|-------------|------------------------|
| 0/100 (seed 42)        | 8.359                  | 86.76         | 1.588       | energy-only collapse   |
| 25/75 (seed 42)        | 0.912                  | 11.39         | 0.173       | energy-weighted usable |
| 50/50 (N=5 mean±std)   | 0.893±0.081            | 13.01±6.62    | 0.187±0.078 | neutral, CV=0.42       |
| 75/25 (N=5 mean±std)   | 0.799±0.100            | 9.23±6.67     | 0.139±0.085 | practical, CV=0.61     |
| 100/0 (seed 42)        | 0.631                  | 1.49          | 0.032       | best seed-42 comfort   |

**Table S21:** Per-seed MORL yearly metrics for the two canonical weight pairs.

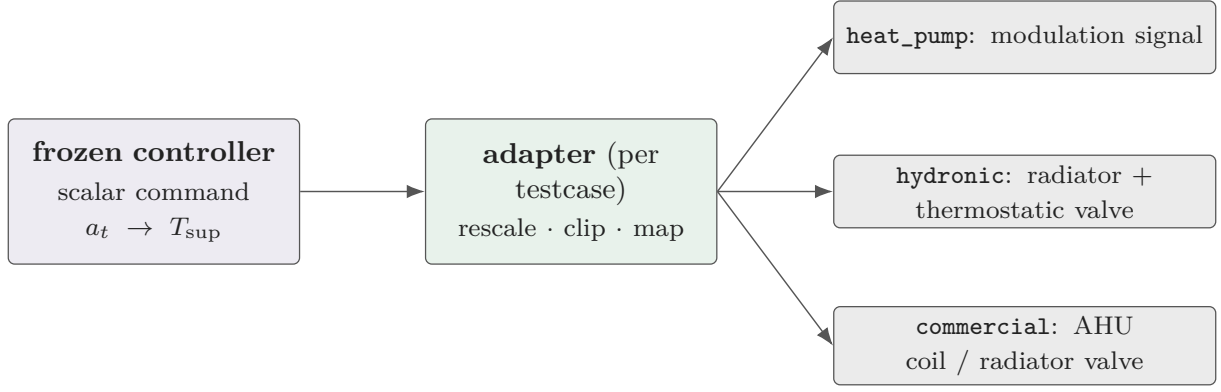
| Pair (c/e) | Seed | RMSE <sub>T</sub> | Within 1°C (%) | Violation (%) | Energy (kWh) | $m_s$ |
|------------|------|-------------------|----------------|---------------|--------------|-------|
| 50/50      | 42   | 0.939             | 74.9           | 12.92         | 2843.6       | 0.193 |
| 50/50      | 43   | 0.769             | 82.9           | 6.77          | 2872.0       | 0.103 |
| 50/50      | 44   | 0.879             | 76.9           | 9.34          | 2858.4       | 0.141 |
| 50/50      | 45   | 0.895             | 77.7           | 11.97         | 2839.8       | 0.187 |
| 50/50      | 46   | 0.985             | 65.4           | 24.05         | 2554.1       | 0.310 |
| 75/25      | 42   | 0.700             | 83.3           | 3.08          | 2994.3       | 0.057 |
| 75/25      | 43   | 0.716             | 84.9           | 5.47          | 2963.9       | 0.087 |
| 75/25      | 44   | 0.769             | 81.7           | 7.82          | 2819.1       | 0.117 |
| 75/25      | 45   | 0.903             | 78.5           | 9.42          | 3036.7       | 0.160 |
| 75/25      | 46   | 0.909             | 70.3           | 20.37         | 2602.8       | 0.276 |



*the manifest is git-tagged before any evaluation (audit anchor) — no post-hoc threshold tuning*

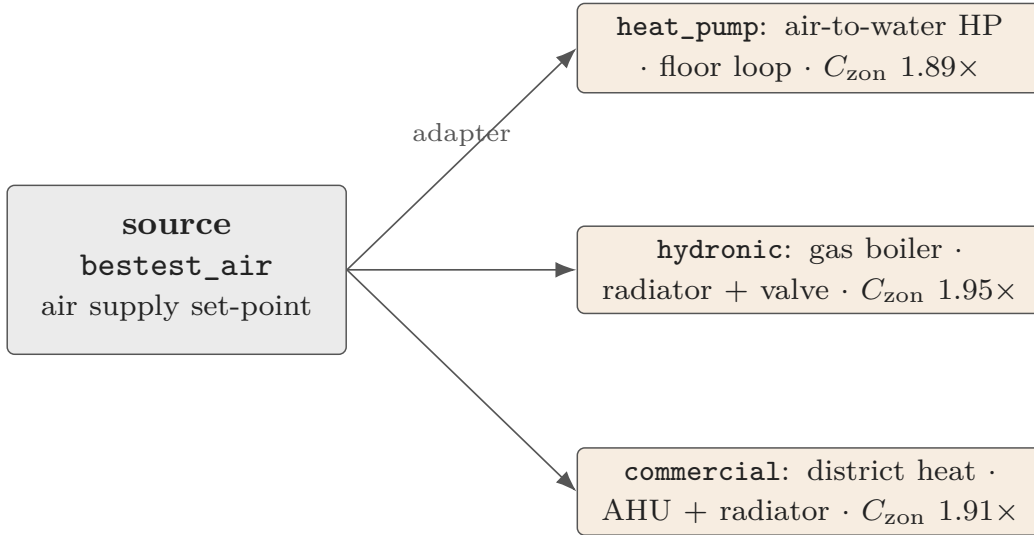
**Figure S19:** Pre-specified transferability protocol. Controller-side transfer is tested against a testcase-specific PI threshold; surrogate-side transfer is tested by full Stage A/B/C recalibration on target telemetry.





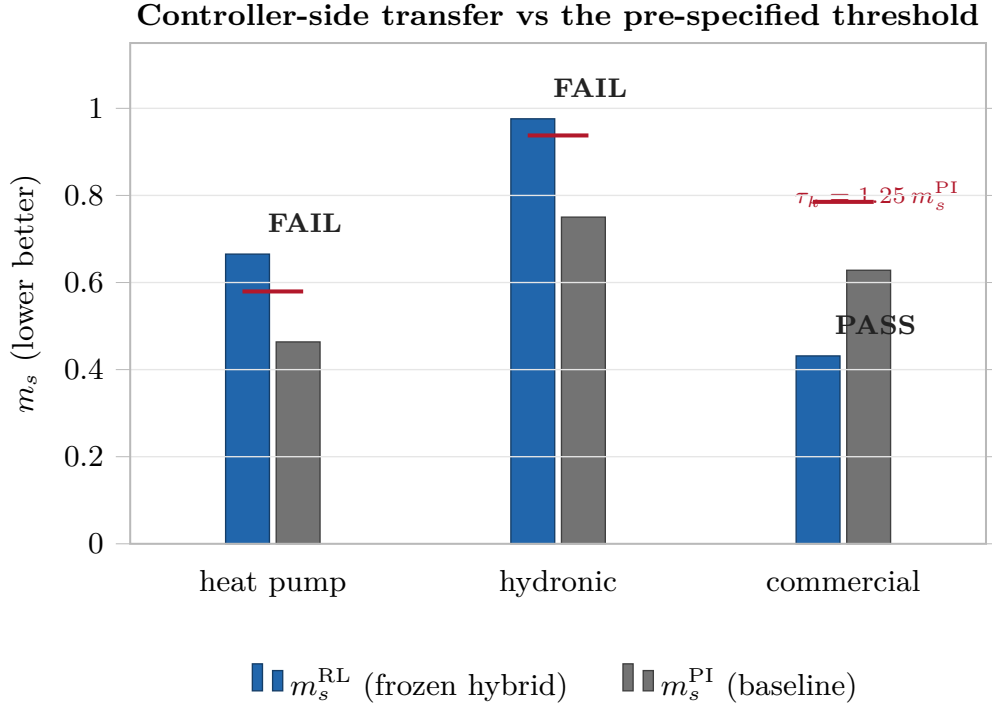
*Transfer is adapter-mediated: the frozen policy’s scalar set-point is re-mapped to each test case’s actuator interface — not a literal copy of the source actions.*

**Figure S20:** Adapter-mediated transfer schematic. The frozen hybrid policy is routed through a pre-specified actuator adapter  $\mathcal{A}_k$  to each testcase’s actuator interface; the controller-side verdict (against the  $1.25\times$  PI threshold) is per-testcase, while full Stage A/B/C surrogate recalibration passes on all three.

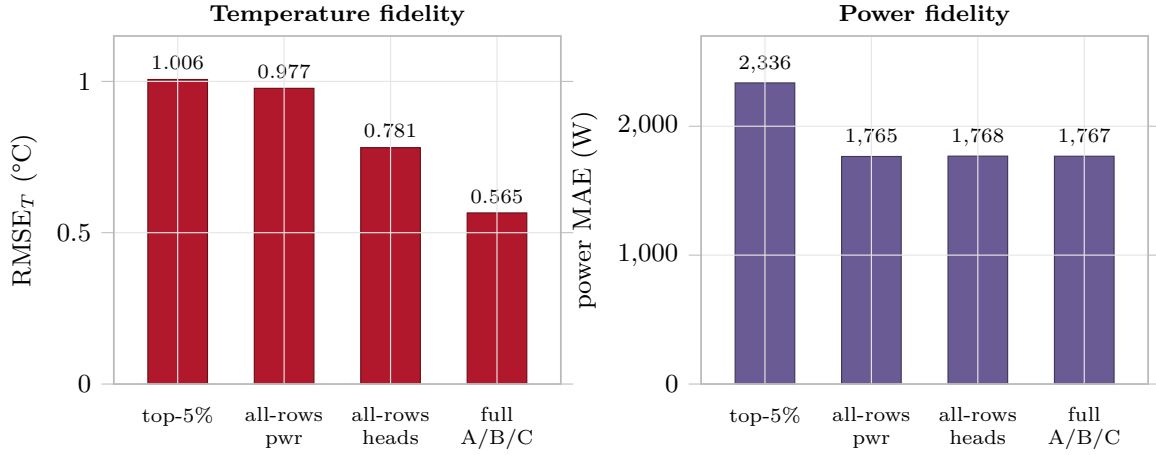


*the hydronic family shares a near-constant  $C_{zon}$  ratio ( $\approx 1.9\times$ ), a family regularity exploited for transfer.*

**Figure S21:** Schematic (no data). Comparative HVAC topology of the source case and the three hydronic targets (source  $\rightarrow$  distribution  $\rightarrow$  zone). Transfer changes the heat source, distribution path, and actuator set, while the single-zone energy balance and the zone thermal capacitance  $C_{zon}$  are shared across the family.



**Figure S22:** Controller-side transfer: frozen-hybrid  $m_{s,\text{RL}}$  against the PI baseline  $m_{s,\text{PI}}$  and the pre-specified threshold  $\tau = 1.25 m_{s,\text{PI}}$  (dashed). The two residential cases exceed the threshold (FAIL); the commercial case is below it (PASS, with the separate energy caveat).



**Surrogate fidelity improves with recalibration depth (controller frozen)**

**Figure S23:** Primary-testcase surrogate-fidelity progression across recalibration regimes. Temperature and power fidelity improve monotonically toward full Stage A/B/C, while the frozen controller's live  $m_s$  is invariant by experimental design.

**Hypothesis closure vs version-locked predictions**

| Hypothesis / prediction | Observed closure                      | Evidence                       |
|-------------------------|---------------------------------------|--------------------------------|
| ● H1 strong             | FALSIFIED                             | frozen recipe, direct transfer |
| ● H2 medium             | FALSIFIED (by structural definition)  | partial recalibration          |
| ● H3 surrogate          | SUPPORTED (on N=3)                    | full Stage A/B/C RMSE          |
| ● H3 controller         | FALSIFIED (regime-dependent failure)  | controller full regime         |
| ● Stretch none          | expected FAIL → observed PASS         | commercial, mode=none          |
| ● Stretch $C_{zon}$     | expected scale-dep. → observed 1.909× | commercial $C_{zon}$           |
| ● Stretch RMSE          | expected 50–90% → observed 87.81%     | commercial recalibration       |

**Figure S24:** Hypothesis closure matrix: the transfer-boundary experiment supports surrogate-side transferability but falsifies deployment-ready frozen-controller transfer.

**Table S22:** Pre-specified target testcases and actuator adapters.

| Role      | Testcase                              | Structural difference vs <b>bestest_air</b>                | Adapter                             |
|-----------|---------------------------------------|------------------------------------------------------------|-------------------------------------|
| Primary   | <b>bestest_hydronic_heat_pump</b>     | hydronic loop driven by a heat pump                        | <b>hydronic_t_supply_v1</b>         |
| Secondary | <b>bestest_hydronic</b>               | boiler/radiator hydronic distribution                      | <b>hydronic_direct_supply_v1</b>    |
| Stretch   | <b>singlezone_commercial_hydronic</b> | order-of-magnitude larger commercial zone + hydronic valve | <b>commercial_hydronic_valve_v1</b> |

**Table S23:** Per-testcase actuator-adapter mapping. All adapters share  $h = \text{clip}((T_{\text{sup},t} - 18)/17, 0, 1)$ ; setpoint overrides are in kelvin.

| Testcase          | Primary override              | Override formula                              | Auxiliary overrides                                              |
|-------------------|-------------------------------|-----------------------------------------------|------------------------------------------------------------------|
| <b>heat_pump</b>  | <b>oveTSet_u</b> (K)          | $288.15 + 9h$ (zone setpoint 15–24°C)         | <b>oveHeaPumY_u</b> , <b>ovePum_u</b> , <b>oveFan_u</b> from $h$ |
| <b>hydronic</b>   | <b>oveTSetSup_u</b> (K)       | $T_{\text{sup},t} + 273.15$ (direct supply)   | <b>ovePum_u</b> from $h$ ; fixed comfort-band context            |
| <b>commercial</b> | <b>dh_oveTSupSetHea_u</b> (K) | $T_{\text{sup},t} + 273.15$ (district supply) | heating valves + pump from $h$ ; fixed zone/air context          |

**Table S24:** Pre-specified recalibration regimes. The controller is frozen in every regime.

| Regime  | Surrogate action                                | Controller action        |
|---------|-------------------------------------------------|--------------------------|
| none    | frozen <b>bestest_air</b> surrogate recipe      | frozen hybrid controller |
| partial | Stage C residual-head recalibration only        | frozen hybrid controller |
| full    | Stage A + Stage B + Stage C on target telemetry | frozen hybrid controller |

**Table S25:** Testcase architecture and identified physics.  $C_{\text{zon}}$  in  $10^5$  J/K; equivalent mass =  $C_{\text{zon}}/c_{p,\text{air}}$ ; mean delivered power from the Stage-C telemetry.

| Testcase                    | Heat source     | $C_{\text{zon}}$ | eq. mass (kg) | mean power (W) | Thermal regime            |
|-----------------------------|-----------------|------------------|---------------|----------------|---------------------------|
| <b>bestest_air</b> (source) | air supply      | 4.413            | 439           | —              | reference                 |
| hydronic heat pump          | heat pump       | 8.348            | 831           | 2,778          | conduction-dominated      |
| hydronic (boiler)           | boiler/radiator | 8.623            | 858           | 679            | slow hydronic             |
| commercial hydronic         | district + AHU  | 8.426            | 838           | 76,860         | ventilation/AHU-dominated |

**Table S26:** Primary-testcase per-regime detail.  $m_s$  is invariant across rows because the controller is frozen by experimental design; only surrogate fidelity changes.

| Regime / result              | RMSE <sub>T</sub> (°C) | Power MAE (W) | $m_{s,\text{RL}}$ | Status                    |
|------------------------------|------------------------|---------------|-------------------|---------------------------|
| PI baseline (yearly)         | —                      | —             | 0.464             | reference                 |
| none: frozen controller      | —                      | —             | 0.665             | control FAIL              |
| partial: Stage C top-5%      | 1.006                  | 2336          | 0.665             | diagnostic                |
| partial: all-rows power head | 0.977                  | 1765          | 0.665             | diagnostic (power)        |
| partial: all-rows heads      | 0.781                  | 1768          | 0.665             | surrogate cond. pass      |
| full: Stage A/B/C            | 0.565                  | 1767          | 0.665             | surrogate PASS, ctrl FAIL |

**Table S27:**  $C_{\text{zon}}$  re-identification across the **bestest\_air** baseline and the three hydronic testcases.

| Testcase                         | $C_{\text{zon}}$ (J/K) | $\rho_C$ vs <b>bestest_air</b> |
|----------------------------------|------------------------|--------------------------------|
| <b>bestest_air</b> (source case) | $4.413 \times 10^5$    | 1.000 (baseline)               |
| hydronic heat pump               | $8.348 \times 10^5$    | 1.892                          |
| hydronic (boiler)                | $8.623 \times 10^5$    | 1.954                          |
| commercial hydronic              | $8.426 \times 10^5$    | 1.909                          |
| hydronic-family mean $\pm$ std   | —                      | $1.918 \pm 0.032$              |

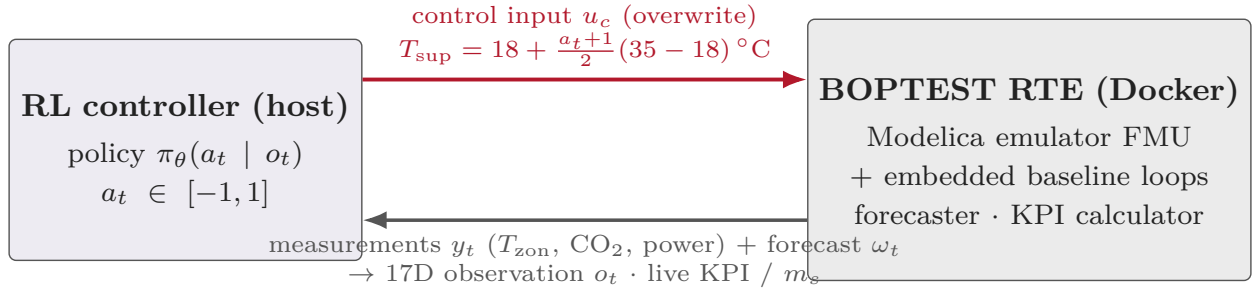
**Table S28:** Stretch-testcase pre-specified predictions versus observed outcomes.

| Prediction                           | Pre-reg.            | A-priori | Observed        | Verdict   |
|--------------------------------------|---------------------|----------|-----------------|-----------|
| mode=none controller verdict         | FAIL                | 0.80     | threshold PASS  | FALSIFIED |
| full surrogate RMSE gain             | [50, 90]%           | 0.70     | 87.8%           | CONFIRMED |
| $C_{\text{zon}}$ hyp. A (uniform)    | [1.7, 2.2] $\times$ | 0.35     | 1.91 $\times$   | CONFIRMED |
| $C_{\text{zon}}$ hyp. B (scale-dep.) | [3, 10] $\times$    | 0.50     | 1.91 $\times$   | FALSIFIED |
| $C_{\text{zon}}$ hyp. C (failure)    | non-convergence     | 0.15     | converged (N=3) | FALSIFIED |

## BOPTEST RTE control loop and supply-temperature actuation

The live evaluation runtime is the BOPTEST Run-Time Environment (RTE): a Dockerised Modelica emulator with embedded baseline local-loop and supervisory controllers, a disturbance forecaster, and a KPI calculator, all exposed through a RESTful HTTP API (Blum et al., 2021). The test controller does not replace the plant; following the BOPTEST formalism it *overwrites* a chosen subset  $u_c$  of the emulator’s control inputs while the remaining embedded loops keep running at baseline. In this study  $u_c$  is the single supply-temperature set-point of each test case, activated through the BOPTEST

overwrite pair (the point’s `_activate` flag set to 1 and its `_u` value written); on `bestest_air` this point is the supply-air-temperature set-point `oveTSetSup`. At every control step  $\Delta t = 900$  s (Option-1 synchronisation, in which the controller defines the step and the emulator advances when an action is returned) the controller reads measurements  $y_t$  (zone temperature, CO<sub>2</sub>, power) and disturbance forecasts  $\omega_t$  from the API, maps its policy output  $a_t \in [-1, 1]$  to the supply-temperature command  $T_{\text{sup}} = 18 + \frac{a_t+1}{2}(35 - 18)^\circ\text{C}$ , writes it to the overwrite signal, and issues `POST /advance` to integrate the emulator forward one step. The RTE then returns the next measurements and the live key performance indicators (including the maintenance score  $m_s$ ) computed from the closed-loop trajectory rather than from the surrogate. For the hydronic transfer cases the same supply-temperature command is routed through the per-test-case adapter to that case’s overwrite signal (heat-pump modulation, valve position, or supply-water set-point), so the frozen policy interface is unchanged across test cases (Figure S25).



one control step  $\Delta t = 900$  s (Option-1 synchronisation via `POST /advance`)

**Figure S25:** BOPTEST RTE closed-loop control interface (topology after Blum et al., 2021, Figs. 1–3). At each 900 s step the policy overwrites only the supply-temperature set-point through the `oveTSetSup` signal pair and advances the Dockerised emulator, which returns measurements, forecasts, and the live maintenance score  $m_s$ .

## Detailed formulas

The following standard or implementation-level equations were moved here to keep the main article focused on results; they are reproduced for completeness. **PPO clipped surrogate objective** (Schulman et al., 2017), with  $\hat{A}_t$  the GAE advantage:

$$L_{\text{CLIP}}(\theta) = \mathbb{E}_t \left[ \min(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right], \quad \rho_t(\theta) = \frac{\pi_\theta(a_t \mid o_t)}{\pi_{\theta_{\text{old}}}(a_t \mid o_t)}. \quad (2)$$

**Black-box (BB) multi-horizon training loss** with physical-consistency penalty:

$$\mathcal{L}_{\text{BB}} = \sum_{k \in \{1, 2, 4\}} \frac{1}{N} \sum_i (\hat{T}_i(t+k) - T_i(t+k))^2 + \lambda_{\text{phys}} \text{ReLU}(|\Delta \hat{T}| - \Delta T_{\text{max}}). \quad (3)$$

**Grey-box (GB) positive reparameterization** of the zone capacitance:

$$C_{\text{zon}} = c_s(\text{softplus}(\rho) + c_{\text{min}}/c_s), \quad C_{\text{zon}} \geq c_{\text{min}} = 5 \times 10^4 \text{ J/K}. \quad (4)$$

**Stage B maximum-a-posteriori identification** of  $C_{\text{zon}}$  on the excitation subset  $\mathcal{E}$  under a weak Gaussian prior:

$$\hat{C}_{\text{zon}} = \arg \min_C \sum_{i \in \mathcal{E}} (\hat{T}_i(C) - T_i)^2 + \frac{1}{2\sigma_p^2} (C - C_{\text{prior}})^2. \quad (5)$$

**Warm-started GB heat-term scaling:** when a GB surrogate is warm-started from legacy BB weights, the inherited heat term is rescaled so that the per-step increment is dimensionally consistent at the runtime control step (the BB update lacks this scaling because its increment carries no explicit  $\Delta t$ ):

$$s = s_{\text{legacy}} \frac{C_{\text{init}}}{\Delta t_{\text{legacy}} q_{\text{scale}}}. \quad (6)$$

**Transfer-gap diagnostics:** surrogate–live maintenance-score mismatch and mean action-gap norm:

$$\begin{aligned} \Delta m_s &= m_{s,\text{surrogate}} - m_{s,\text{BOPTEST}}, \\ g_a &= \frac{1}{N} \sum_{t=1}^N \|a_{t,\text{BOPTEST}} - a_{t,\text{surrogate}}\|_2. \end{aligned} \quad (7)$$

## Tables relocated from the main article

The following tables were moved here to reduce the float load of the main article; they are cited from it as Supplementary Tables.

**Table S29:** Maintenance-score decomposition across controller families (cited in the main article’s controller-family synthesis). Thermostatic/HDRL are 14-day windows; MORL/PI are yearly means.

| Controller (window/eval)          | $m_s$ | $r_{\text{time}}$ | $r_{\text{sev}}$ |
|-----------------------------------|-------|-------------------|------------------|
| pure BB (peak)                    | 0.073 | 0.015             | 0.058            |
| direct GB (peak)                  | 1.046 | 0.771             | 0.276            |
| hybrid $\lambda_T=0.10$ (peak)    | 0.087 | 0.047             | 0.040            |
| HDRL $\lambda_T=0$ (peak)         | 0.180 | 0.061             | 0.119            |
| pure BB (typical)                 | 0.095 | 0.044             | 0.051            |
| direct GB (typical)               | 1.102 | 0.824             | 0.278            |
| hybrid $\lambda_T=0.10$ (typical) | 0.041 | 0.024             | 0.017            |
| HDRL $\lambda_T=0$ (typical)      | 0.234 | 0.031             | 0.202            |
| MORL 50/50 (yearly, N=5)          | 0.187 | 0.130             | 0.057            |
| MORL 75/25 (yearly, N=5)          | 0.139 | 0.092             | 0.047            |
| PI (yearly)                       | 0.910 | 0.636             | 0.274            |

**Table S30:** Controller-learning falsifiable-hypothesis ledger (CL1–CL5), cited in the main article’s central fidelity–utility test. These are distinct from the paper-level hypotheses H1–H4: CL1–CL3 are the controller-experiment counterparts of H1–H3, and CL4–CL5 add MORL-specific checks. Each hypothesis is tested on the live BOPTEST RTE.

| Hypothesis                                                                                                  | Evidence      | Verdict                                               |
|-------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------|
| CL1 The higher-fidelity twin (GB) used directly as the rollout environment yields the best controller.      | central test  | FALSIFIED (direct GB $m_s > 1.0$ )                    |
| CL2 Calibrated GB is useful when its role changes from dynamics provider to a frozen reward-shaping censor. | hybrid remedy | SUPPORTED (hybrid gives best cross-window robustness) |
| CL3 The thermostatic-optimal censor weight $\lambda_T = 0.10$ transfers to the HDRL family.                 | HDRL sweep    | FALSIFIED (HDRL best at $\lambda_T = 0$ )             |
| CL4 MORL viability is determined by the reward scalarization alone.                                         | MORL 5D/17D   | FALSIFIED (needs the 17D observation)                 |
| CL5 The single-seed MORL canonical ( $m_s \approx 0.10$ ) is representative.                                | MORL N=5      | FALSIFIED at N=5 (best of five; CV 0.42–0.61)         |

**Table S31:** Modelling assumptions of the direct supply-temperature ( $T_{\text{sup}}$ ) control interface (cited in the main article’s predictive-validation section).

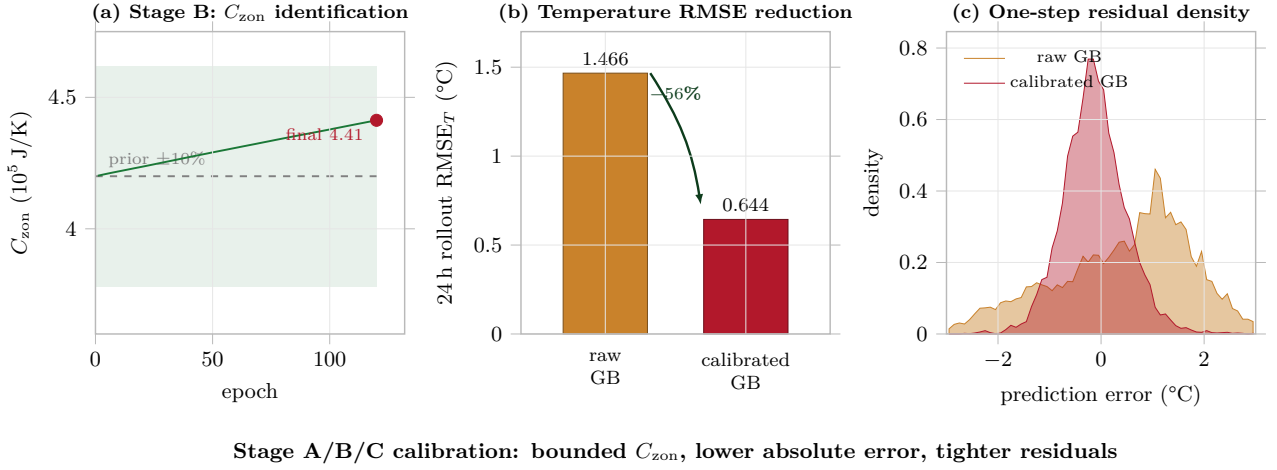
| Assumption           | Statement                                                                                                           |
|----------------------|---------------------------------------------------------------------------------------------------------------------|
| Actuator abstraction | Supply-air temperature is commanded directly; valve/coil dynamics are not modelled beyond the Stage A latency term. |
| Command bounds       | $T_{\text{sup}} \in [18, 35]^\circ\text{C}$ ; fan command $u_{\text{fan}} \in [0, 1]$ .                             |
| Power non-negativity | $\hat{P} = \text{Softplus}(\cdot) \geq 0$ , capped at $P_{\text{max}} = 5500$ W.                                    |
| State support        | Zone and ambient temperatures normalized to $[-1, 1]$ over $[15, 35]$ and $[-10, 40]^\circ\text{C}$ .               |
| Interface invariance | All four backends share the same transition signature; only the internal dynamics differ.                           |

**Table S32:** Transition-dataset schema for the surrogate models. Every field of a training tuple  $(s_t, a_t, d_t, s_{t+1})$  is tagged by role: **M** BOPTEST measurement, **X** exogenous disturbance, **D** derived feature, **A** control action, **T** regression target. In a recursive rollout the predicted targets update the physical-state block of the next observation ( $\hat{T}_{\text{zone},t+1} = T_{\text{zone},t} + \Delta\hat{T}_{\text{zone},t+1}$ , power, previous action), while exogenous, forecast, and time fields advance from the disturbance stream; the policy then emits  $a_{t+1}$  and the map  $f_\theta$  is re-applied.

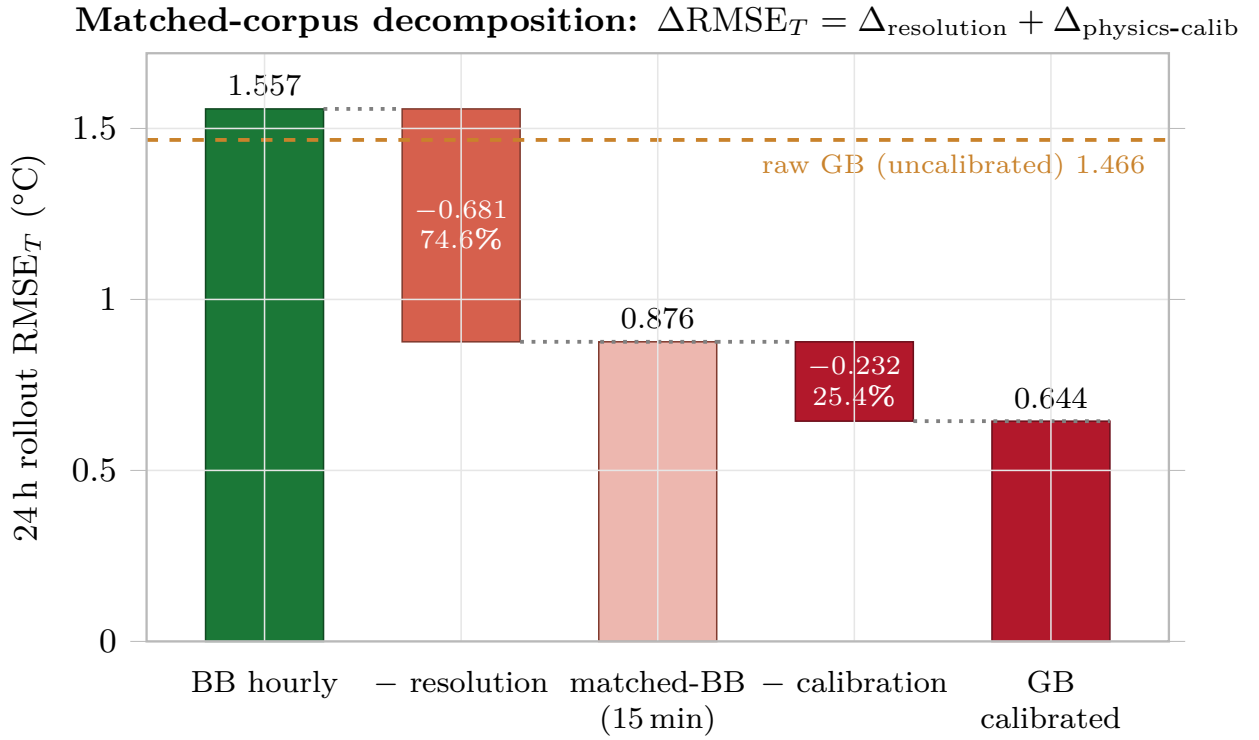
| Field                                                               | Role | Description                                                                                            |
|---------------------------------------------------------------------|------|--------------------------------------------------------------------------------------------------------|
| $T_{\text{zone},t}$                                                 | M    | zone air temperature (BOPTEST sensor); in the observation and, via $\Delta T$ , in the target          |
| $P_t$ (HVAC power)                                                  | M    | measured heating/cooling power; observation and power-head target                                      |
| $\text{CO}_2$                                                       | M    | zone $\text{CO}_2$ concentration; observation only                                                     |
| $T_{\text{amb},t}$                                                  | X    | ambient temperature (weather); observation                                                             |
| solar gain, occupancy, comfort set-points                           | X    | exogenous drivers in $d_t$ ; affect the dynamics, not all directly observed                            |
| $\hat{T}_{\text{amb},t+1:t+24\text{h}}$                             | D    | 5 ambient forecasts (+1, +3, +6, +12, +24 h); observation                                              |
| cyclic time features                                                | D    | hour/day sine-cosine encodings; observation                                                            |
| prev. supply temp. $T_{\text{sup},t-1}$ ,<br>prev. action $a_{t-1}$ | D    | control history; observation                                                                           |
| $\Delta T_{\text{zone},t}$                                          | D    | causal-smoothed temperature delta; observation                                                         |
| $a_t = a_0$ (supply-T)                                              | A    | commanded supply-temperature set-point (main-text action map); the only deployed control input         |
| $a_1$ (fan)                                                         | A    | fan command; excited only during offline corpus generation, held at the BOPTEST baseline at deployment |
| $\Delta\hat{T}_{\text{zone},t+1}$                                   | T    | predicted next-step zone-temperature increment                                                         |
| $\hat{P}_{t+1}$                                                     | T    | predicted next-step HVAC power (power head)                                                            |
| internal surrogate state (GB: RC state / $C_{\text{zon}}$ )         | —    | latent; propagated by the RC balance (GB) or carried within $s_t$ (BB)                                 |

**Table S33:** Staged inverse calibration of the GB surrogate under the `block1_15min_episodeaware` preset; a second `power_head_only` preset freezes  $C_{\text{zon}}$  and refines only the power residual head (80 epochs).

| Stage | Calibrates                                                                                                | Key result                                                                                                                                                                                              |
|-------|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A     | telemetry alignment: one-step latency & bias                                                              | bias $-0.105^\circ\text{C}$ ; post-fit temperature RMSE $0.238^\circ\text{C}$                                                                                                                           |
| B     | zone capacitance $C_{\text{zon}}$ on the top-excitation subset (403/8058 rows, quantile 0.95; 120 epochs) | $C_{\text{zon}} = 4.413 \times 10^5$ J/K (+5.1% over prior); last-20-epoch stability $\pm 1,005$ J/K (0.23%); Laplace/Fisher $1\sigma = 6.6\%$ , 95% CI $[3.845, 4.980] \times 10^5$ J/K ( $n = 1838$ ) |
| C     | neural residual heads, $C_{\text{zon}}$ frozen (60 epochs)                                                | refines residual fit while preserving the physical parameter                                                                                                                                            |

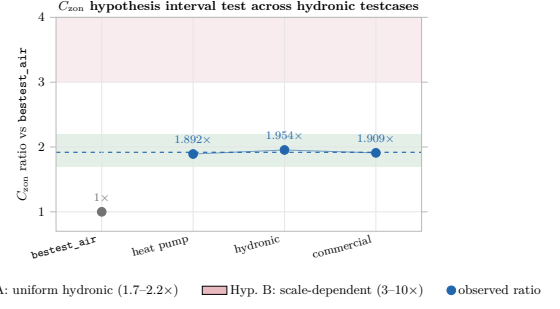
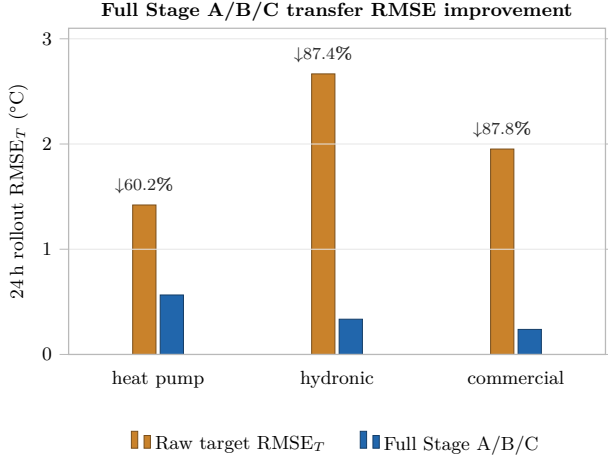


**Figure S26:** Stage A/B/C diagnostics:  $C_{zon}$  convergence, rollout temperature-error reduction, and residual tightening.



**Figure S27:** Matched-corpus attribution. The BB-to-GB gain is split into the 15-minute corpus contribution and the Stage A/B/C calibration contribution.





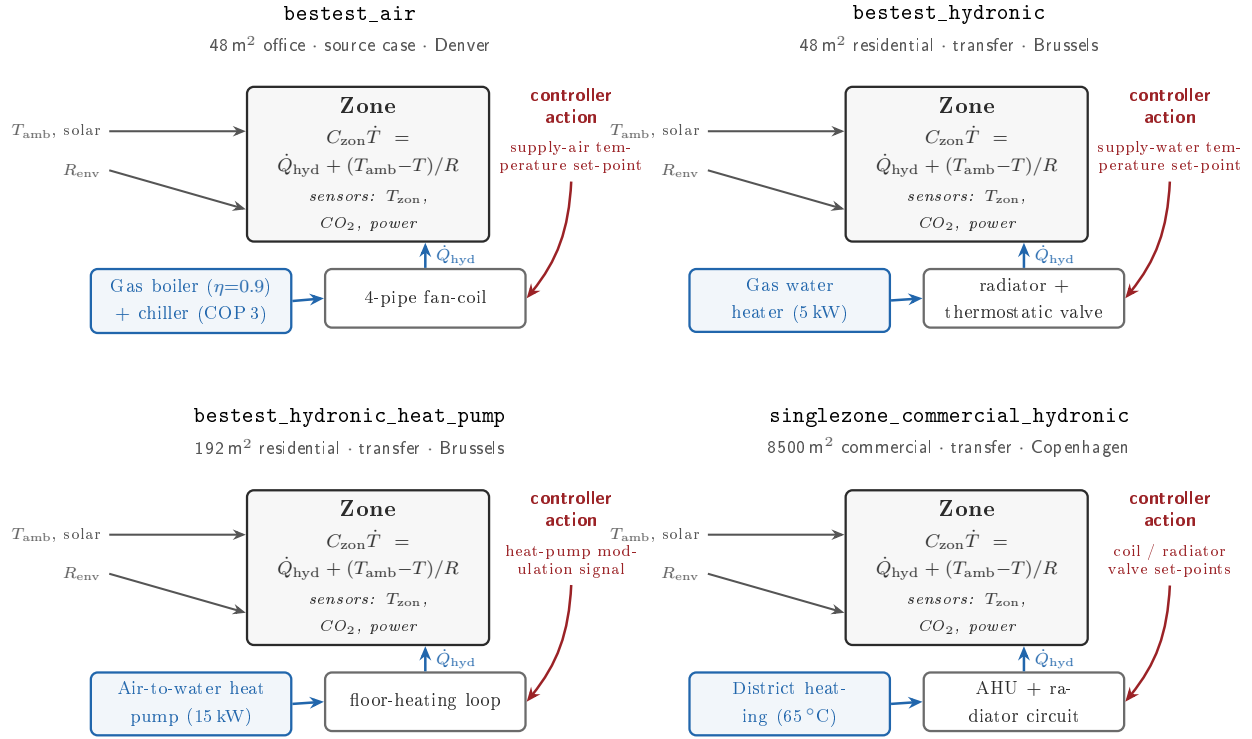
(b)  $C_{zon}$  hypothesis interval.

(a) Full Stage A/B/C RMSE improvement.

**Figure S28:** Surrogate-side transfer and physical consistency. Full Stage A/B/C reduces target rollout error on all hydronic-family testcases, while the re-identified  $C_{zon}$  ratios fall inside the uniform hydronic-family interval and outside the scale-dependent interval.

**Table S34:** Pre-specified hypothesis closure.

| Hypothesis      | Claim                                                    | Verdict                | Evidence                                                                                                          |
|-----------------|----------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------|
| T1 (strong)     | Frozen mode=none recipe gives deployment-ready transfer. | FALSIFIED              | Residential cases fail the 1.25× PI threshold; commercial is threshold-pass only with +35.3% energy.              |
| T2 (medium)     | Partial Stage C recovers controller transfer.            | FALSIFIED (structural) | Partial recalibrates only the surrogate; the controller is frozen by scope, so live KPI cannot change.            |
| T3 (surrogate)  | Full Stage A/B/C gives a usable target surrogate.        | SUPPORTED (N=3)        | RMSE <sub>T</sub> gains 60.2%, 87.4%, 87.8%; $C_{zon}$ re-identified consistently.                                |
| T3 (controller) | Frozen controller transfers given a valid adapter.       | FALSIFIED              | Residential cases fail comfort; commercial passes safety but overuses energy, regime-dependent failure.           |
| hierarchy       | Verdict is monotone in recalibration depth.              | SUPPORTED              | Controller KPI invariant under surrogate-only recalibration; surrogate fidelity improves with full recalibration. |



**Figure S29:** Schematics of the four BOPTEST test cases (source **bestest\_air** plus the three hydronic transfer targets), from the official BOPTEST documentation. All share the single-zone energy balance but differ in heat source, distribution path, and actuator interface; the highlighted arrow marks the scalar command channel or adapter-mediated actuator interface used by the controller in each case.